

SIGN/e

graphic notation

SIGN/e is a Max for Live plug-in for creating and performing animated scores. It allows users to import, move, modify, and animate symbols (PNG), opening singular possibilities for musical creation.

USER MANUAL
Version 1.0 • May 15th, 2026

Quick Start

1. Download and install the plug-in:
<https://lfo-lab.ca/logiciel>
2. Drag SIGNe-Screen.amxd ont a new MIDI track
A video window appears
3. Drag SIGNe-Symbol.amxd ont a new MIDI track
A circle appears in the video screen
4. Start working: change symbol parameters using automation and animate your score!

Shortcuts

Function	Mac	Windows
Move Symbol	Click a symbol, then drag	Left click a symbol, then drag
Select multiple symbols	Click & drag over symbols	Click & drag over symbols
Deselect Symbol	⌘ (command) + click & drag	Ctrl + click & drag
Resize symbol	⌥ (option) + click & drag	Alt + click & drag
Rotate symbol	⇧ (shift) + click & vertical drag	Shift + click & vertical drag
Symbol opacity	O + click & vertical drag	O + click & vertical drag
Mouse Scroll	T + click & horizontal drag	T + click & horizontal drag
Full screen (enter and exit)	ESC	ESC

Adding and Creating Symbols

You must add one SIGNe-Symbol.amxd track for each symbol.

Selection is simple: hover and click. A yellow bounding box appears.

Multiple objects can be selected using click-drag selection.

To replace the default symbol, click **Properties** in SIGNe-Symbol.amxd. In the Symbol section, choose a symbol. Libraries can be navigated via dropdown menus and arrows.

Importing Custom Symbols

To import your own symbols (created in external graphic software), drag your images into symbols/ O_Default/User.

Images will appear in SIGNe-Symbol.amxd → Properties → Symbol → O_Default → User. Your images should respect the following parameters :

Recommended format and resolution:

- PNG
- 300 DPI
- 500 x 500 pixels
- Transparent background
- White symbol

Exporting your images in white is recommended so color can be adjusted in directly in SIGNe-Symbol.amxd.

High resolution ensures better display quality. Very small images may produce artifacts.

If using many symbols, reducing resolution (150 DPI, 250 × 250 px) improves performance.

Setting up a score

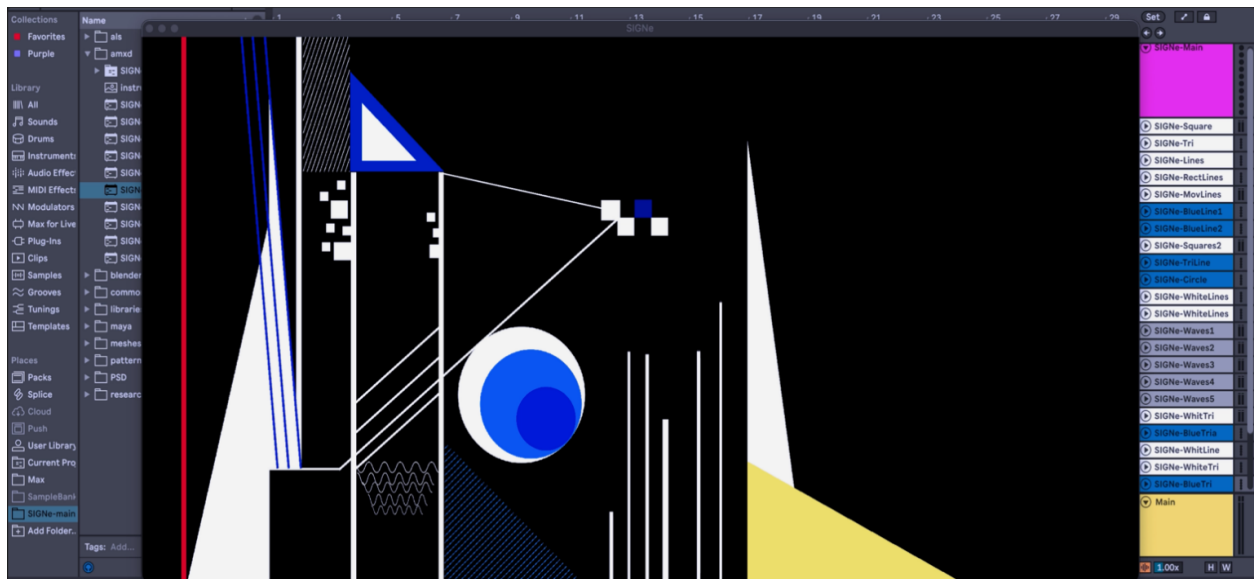
To begin, add one SIGNe-Screen.amxd instance. It acts as the main compiler for all other instances and creates the video window.

- 1) **Drag SIGNe-Screen.amxd** into a MIDI track. A video screen appears
- 2) **Drag SIGNe-Symbol.amxd** into another MIDI track. A symbol appears.
- 3) To replace the default graphic element, click on Properties in the SIGNe-Symbol.amxd and navigate to your preferred library using the drop-down menus. You can click on the graphic element of your choice, and it will replace the previous one.

Repeat step 3 for each new symbol.

IMPORTANT

Only one SIGNe-Symbol.amxd per track is recommended for easier automation management.



The image above shows the very first graphic score created with SIGN/e in November 2025 by Alexandre Sasset-Blouin.

Installation

Mac Installation

- 1) Download installer: <https://lfo-lab.ca/logiciel>
- 2) Unzip the folder. It is important to always keep the folder structure intact so that the links to the graphic element libraries provided with SIGN/e remain active.
- 3) Add SIGNe folder to Ableton Live's browser using "Add Folder"

It is recommended to add the folder using Ableton Live's "Add Folder" function. If the .amxd files are imported, Ableton creates a copy in an internal folder, which breaks the folder hierarchy required for proper operation.

Windows Installation - Additional step

To use the VIDDLL package with Max on a Windows computer, it must be installed via the built-in package manager.

1. Open Max or Max for Live
2. In the menu, click on File > Show Package Manager.
3. Search for VIDDLL with the search bar
4. Install package

The latest version of VIDDLL is not available on Max 8. Use the newest available version.

N.B. Performance on Windows is currently not completely stable on some machines. Please refer to the Windows troubleshooting suggestions section (p. 22).

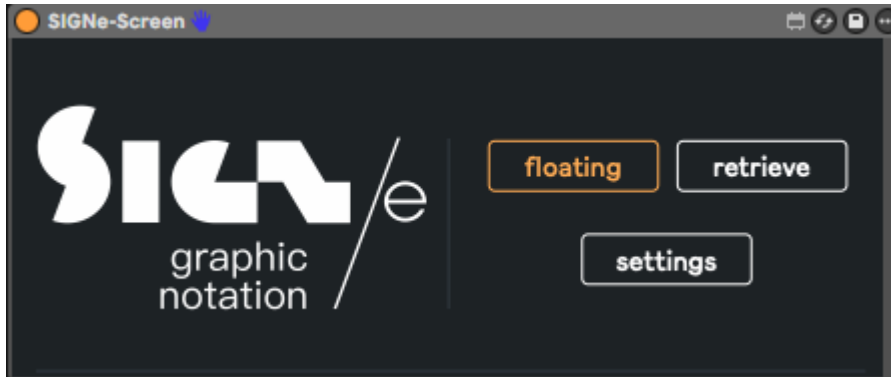
ATTENTION

When an Ableton Live project containing SIGNe objects is loaded, full loading may take several minutes.



Device Descriptions

SIGNe-SCREEN.amxd



Displays and edits the score. Shows all SIGNe-Symbol.amxd instances.

Floating

Brings score window to front.

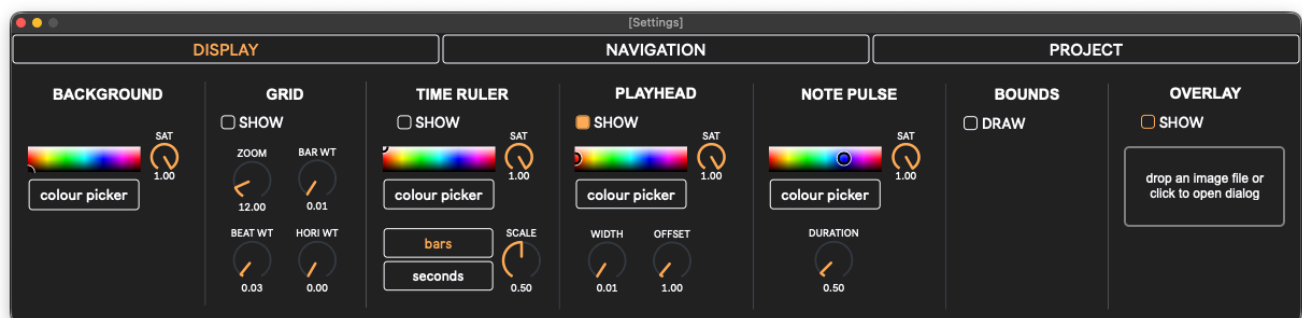
Retrieve

Recalls score window and resets window position.

Settings

Click on settings to open window parameters. The settings are divided in three sections: Display, Navigation et Project.

Settings: DISPLAY



Display: BACKGROUND

Background color and saturation control.

Display: GRID

Shows a rhythmic grid, divided into bars and beats, to help place symbols.

- SHOW: show/hide grid.
- ZOOM: controls the magnification level of the score. This function allows the score to be zoomed in or out without changing the temporal position of its elements. This enables several operations such as selecting multiple elements to move them, taking a screenshot of the entire score, or obtaining an overview.
- BAR WT: controls the width of the bar lines.
- BEAT WT: control the width of the beat lines.
- HORI WT: controls the height of the horizontal lines.

Display: TIME RULER

Shows the Ableton Live transport divisions at the bottom of the score window.

- SHOW: show/hide the time ruler.
- bars/seconds: switch between musical divisions (bars and beats) and time divisions (minutes, seconds).
- SCALE: controls the display size of the time ruler.

Display: PLAYHEAD

Displays an overlaid scrolling line on the video screen for synchronizing events to be performed.

- SHOW: show/hide the playhead.
- WIDTH: controls the width of the playhead.
- OFFSET: controls the playhead's horizontal position.

Display: NOTE PULSE

Settings and color of the pulse when the scrolling line intersects an object.

- DURATION: controls the pulse duration.

Display: BOUNDS

Displays the selection outlines of graphical objects placed on the score.

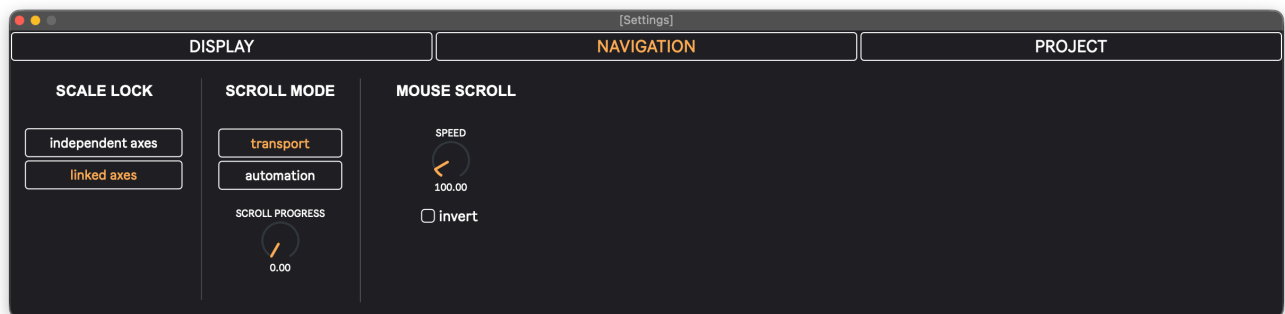
- DRAW: show/hide bounds.

Display: OVERLAY

Allows a PNG image to be overlaid at the beginning of the score, which will scroll along with it. This image is typically used to indicate the instruments being used when the score is horizontally divided by instrument, like a conventional score. This fixed overlay ensures that visual information remains visible even as the score scrolls to the right. It is recommended to keep this image in the same directory as the Ableton Live session.

- SHOW: show/hide the overlay.

Settings: NAVIGATION



Navigation: SCALE LOCK

independent axes: object X and Y scale are unlinked.

linked axes: object X and Y scale are linked.

Navigation: SCROLL MODE

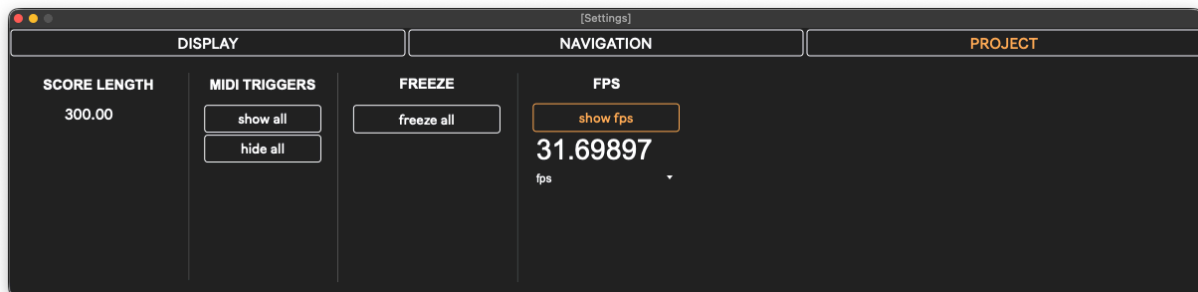
- transport: scrolling is controlled by the Ableton Live transport.
- automation: scrolling is controlled by an automation track automating the SCROLL PROGRESS knob.
- SCROLL PROGRESS: automated or manual scrolling control.

Navigation: MOUSE SCROLL

The mouse scroll function allows navigation of the screen using the mouse. Press T + click and drag left or right.

- invert: invert the scroll direction.
- SPEED: control the scroll speed.

Settings: PROJECT



Project: SCORE LENGTH

Total score length in bars.

Project : MIDI TRIGGERS

- show all: displays a line at the MIDI trigger point of the objects.
- hide all: hide the MIDI trigger lines.

See the **Using MIDI** (p.21) section for more details on MIDI note functions.

Project: FREEZE

Clicking Freeze All locks editing, via the video window, of all objects in the score.

Project: FPS

- Show fps: show/hide FPS performance indicator.

SIGNe-SYMBOL.amxd



This instance allows graphical elements displayed in SIGNe-Screen.amxd to be configured. The parameters directly accessible in the SIGNe-Symbol.amxd window can be controlled via automation or through modulators such as LFO.amxd.

N.B. Automation is less demanding on system resources than modulation and should be prioritized when multiple elements need to be animated.

WARNING

We recommend placing only one SIGNe-Symbol.amxd per track to facilitate identification.

POSITION

POS X / POS Y: move the symbol horizontally or vertically.

ROTATE: rotate the symbol. Can also be done directly on the screen with the shortcut: SHIFT + click & drag.

TRANSFORM

LAYER: layer selection for the symbol. Symbols on higher layers appear above those on lower layers. Layers are numbered from 1 to 10.

OPACITY: controls symbol opacity.

Scale X, Y: adjusts the horizontal and vertical size of the symbol. This can also be done directly on the screen using the shortcut: OPTION + click & drag.

 links or unlinks X and Y size. When activated the size changes are proportional.

N.B. X and Y size parameters must be unlinked when they are being modulated or automated, to avoid operational errors.

REPETITION

This function allows multiple copies of a symbol to be created sequentially, facilitating rhythmic structures. Each repeated element generates the same MIDI note.

When SNAP TO GRID is enabled for the symbol, it will repeat according to the rhythmic grid.

QTY: controls the number of repeated symbols.

SPACIN: controls the spacing between repeated symbols.

ROTATE : controls the repetition axis.

NAVIGATION



Move transport to object: moves the Ableton Live transport cursor to the position of the symbol.

N.B.: This function depends on Ableton's song length parameter. Objects placed by SIGN/e are not recognized as actual elements by Ableton. To extend beyond the default timeline length, place a marker or a MIDI/Audio clip near the intended end of your score (it can be moved afterward).



Move object to transport: moves the symbol to the position of the transport cursor.



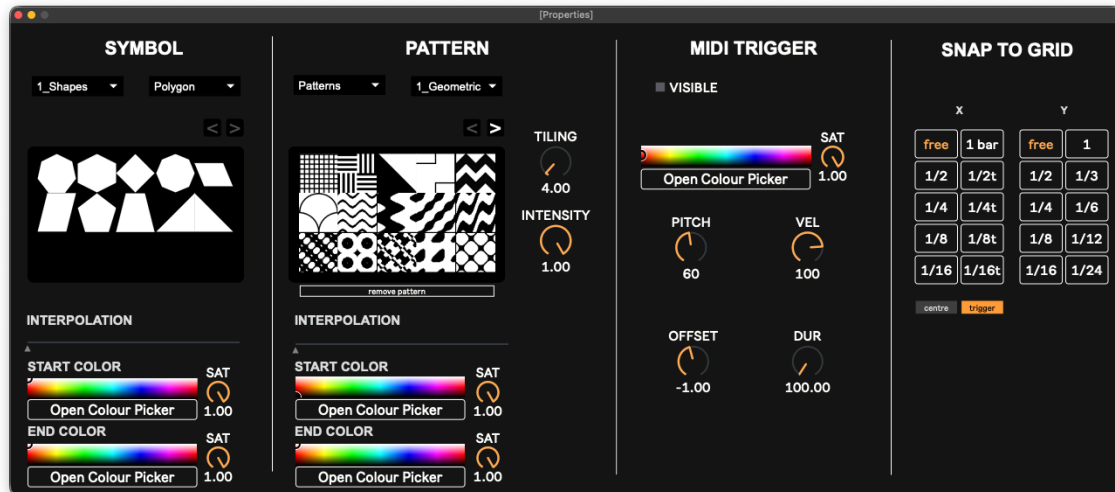
Pin Position: locks the symbol's position on the screen. It will not move as the view scrolls.



Lock Parameters: locks the symbol's positioning and sizing parameters.

Properties

Click on Properties to open the symbol's image, color, pattern, and MIDI settings. The parameters are divided into four sections.



SYMBOL

The Symbol section allows you to change the PNG image of the symbol and its color. See the **Add and Create Symbols** section (p.3) for more details on importing your own symbols.

The desired symbol can be selected using the two drop-down menus. Symbols are organized into libraries, accessible via the left drop-down menu. These libraries can be subdivided into banks, accessible via the right drop-down menu. The structure of the libraries and banks mirrors the folder structure of SIGN/e's library directory.

Bank pages can be navigated using the arrow buttons. Once a symbol is selected, clicking it will load the new symbol onto the screen while preserving the previous symbol's parameters.

Below the symbol selector are the color parameters. It is possible to assign a start and end color and saturation (SAT), which can be interpolated using the Interpolation knob. This interpolation parameter is available for automation.

PATTERN

The Pattern section allows a pattern to be assigned to the symbol. Navigation and selection of patterns work in the same way as symbol selection. Once the desired pattern is chosen, clicking it will apply it to the symbol.

TILING controls the size of the pattern. The higher the value, the smaller the pattern becomes. If the pattern is smaller than the symbol, it will repeat.

INTENSITY controls the opacity of the pattern.

The Remove pattern button removes the pattern. Pattern color selection works in the same way as for the symbol.

MIDI TRIGGER

This section allows you to manage the MIDI information generated by the symbols. By default, the MIDI track where SIGNe-screen.amxd is placed sends a MIDI note every time the Playhead line reaches a symbol's MIDI trigger point. The parameters of the sent note can be modified.

- **VISIBLE:** toggles the visibility of the MIDI trigger line. This line shows precisely where the playhead will trigger a MIDI note.
- **Color:** changes the color of the trigger line.
- **PITCH:** sets the pitch of the MIDI note (MIDI note number).
- **VEL:** sets the velocity of the MIDI note.
- **OFFSET:** delays or advances the MIDI output relative to the symbol's trigger point.
- **DUR:** sets the duration of the MIDI note.
- **WINDOW:** sets the trigger window duration for the MIDI note (the period during which the system checks whether the playhead intersects the MIDI trigger point). Increase this value if some MIDI notes are not being triggered.

SNAP TO GRID

Allows you to select a quantization level for the symbol. When set to free, the symbol is completely unquantized, otherwise, it will snap to the nearest selected division. This enables the creation of rhythms quantized to Ableton's metronome. X-axis divisions follow Ableton's transport bars. Y-axis divisions depend on the Grid, which can be enabled via the Display panel in the Settings of SIGNe-Screen.amxd (see p. 12).

SIGNe-Text.amxd



This instance allows textual elements displayed in SIGNe-Screen.amxd to be configured.

The parameters of SIGNe-Text.amxd operate similarly to those of SIGNe-Symbol.amxd and can also be automated or modulated.

WARNING

We recommend placing a single SIGNe-Text.amxd per track.

Text settings

The leftmost section of the device is used to modify text content, alignment and font.

POSITION

X, Y: move the text horizontally and vertically.

SCALE

Controls the horizontal and vertical size of the text. When  is activated, X and Y sizes change proportionally. Can also be done directly on the score window using the shortcut OPTION + click & drag.

N.B. X and Y size parameters must be unlinked when they are being modulated or automated, to avoid operational errors.

TRANSFORM

LAYER: layer selection for the text.

OPACITY: opacity control for the text.

ROTATE: rotate the text. Can also be done directly in the score window with the shortcut SHIFT + click & drag.

SPACING: controls spacing between characters.

REPETITION

QTY: controls the number of repetitions.

SPACING: controls the spacing between repetitions.

ROTATE: rotates the repetition axis.

BOUNDS: moves the selection frame of the text object. SIGNe attempts to automatically position the frame, but certain fonts may result in a frame that is too small or too large. This function allows manual adjustment. The selection frames of all objects can be displayed in the Display panel available in the Settings of SIGNe-Screen.amxd (see p. 13).

NAVIGATION



Move transport to object: moves the Ableton Live transport cursor to the position of the text.



Move object to transport: moves the text to the Ableton Live transport cursor position.



Pin Position: locks the text's position on the screen. It will not move as the view scrolls.



Lock Parameters: locks the text positioning and sizing parameters.

SIGNe-Background.amxd



This instance allows a background image to be added to SIGNe-Screen.amxd, which will be placed behind symbols and text. This notably allows reading a graphic score that already exists in image form.

Background parameters

drop file: opens the file browser to select a background image.

Clear: removes the background image.

OFFSET

Moves the background in minutes and seconds or in bars and beats.

This function makes it possible to place multiple backgrounds sequentially.

DURATION

Stretches or compresses the background. This parameter is automatically initialized based on the original dimensions of the background. If this initial value is reduced, the score is compressed; if it is increased, the score is stretched. The image will be distorted accordingly. Adjustments can be made in bars or in time (minutes and seconds).

NB: Double-clicking a number resets it to its default value (0). Clicking a number and using the mouse (up/down) or the numeric keypad allows the value to be adjusted.

PRIORITY

Determines whether bars or time is prioritized when the tempo changes.

When BARS is selected, the score length is locked to a fixed number of bars. In the event of a tempo change, its visual length will remain unchanged, but it will scroll faster or slower.

When TIME is selected, the duration defines the score length. If the tempo changes, the score will stretch or compress accordingly so that its length always corresponds to the defined time interval.

SCROLL METHOD

Transport: background scrolling is determined by the Ableton Live transport.

automation: background scrolling is controlled by automation of the SCROLL POSITION knob.

OTHER PARAMETERS

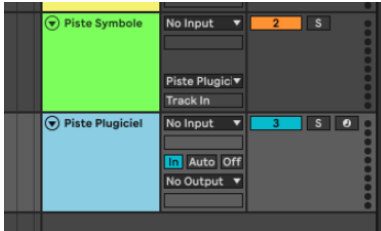
OPACITY: controls the background opacity.

LAYER: layer selection for the background. This function allows multiple backgrounds to be stacked on top of one another.

Background layers are numbered from -10 to -2. This prevents backgrounds from being placed above symbols, which occupy layers 1 to 10. Layer -1 is reserved for the rhythmic grid, and layer 0 is used for temporal divisions.

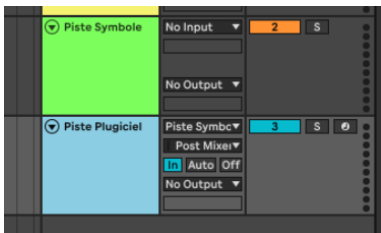
Using MIDI

Symbols placed on the score can send MIDI notes to plugins. Ableton Live provides several options for routing MIDI signals between tracks:



Option 1: Route the output of the MIDI track containing the symbol to the track containing the plugin. This option allows multiple tracks with symbols to be routed to the same plugin.

- Change the **Output Type** of the track containing the symbol to the track containing the instrument plugin.
- Change the **Monitoring** of the track with the instrument plugin to **In**.
- The MIDI from the track containing the symbol will be routed solely to the track with the instrument plugin.



Option 2: sign the input of the track containing the plugin to the output of the track containing the symbol. This option allows the same MIDI information from a track with a symbol to be sent to multiple tracks with different plugins.

- Change the **Input Type** of the MIDI track containing the plugin to the name of the track containing the symbol.
- Change the **Monitoring** of the track with the instrument plugin to **In**.
- The input of the track containing the instrument plugin will now receive MIDI information from the track containing the symbol.
- It is still possible to route other tracks to the track containing the plugin using option 1.

Windows troubleshooting suggestions

The current version of SIGN/e can be unstable on Windows. The following steps can sometimes resolve issues:

1. Delete the files in Max's Database folder. This folder is usually located in:
C:\Users\[username]\AppData\Roaming\Cycling '74\[Max 8 ou Max 9]\Database
2. Open Ableton Live and add an empty Max for Live device (Max MIDI Effect or Max Audio Effect). Click the three dots in the top-right corner of the device. This will recompile the databases.
Delete the files in the Max for Live Devices folder. This folder is usually located in:
C:\Users\[username]\Documents\[Max 8 ou Max 9]\Max for Live Devices
3. Restart the computer.

N.B. The AppData folder is sometimes hidden by default. To access it, go to the folder C:\Users\[username], then click View > Show > Hidden items.

N.B. If the Max for Live Devices folder is not located at this path, check its location by going to Options > File Preferences.

Changelog

You can report any issues here:

<https://github.com/LFO-lab/SIGNe/issues>

License agreement

The license agreement can be found here: <https://github.com/LFO-lab/SIGNe>

Crédits

Direction	Nicolas Bernier
Programming	Evan Montpellier
Product Lead	Pierre-Luc Lecours
Communications & development	Alexandre Sasset-Blouin
Logo	Rutger Zuydervelt

Programmed in 2025/2026 for the Laboratoire formes • ondes of the Digital Music Department of the University of Montreal, as part of the research-creation project *Performative Notation*, supported by the FRQ-SC.

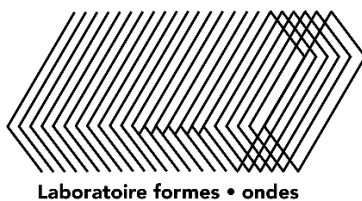
Laboratoire formes • ondes :

<https://lfo-lab.ca>

Digital music programs at the University of Montreal:

<http://musnum.musique.umontreal.ca>

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SIGN/e: Symbol-based Interface for Graphic Notation Edition

A sign is a gesture that conveys information.

A sign allows an idea to be represented.

The artist signs their work.